

Rabiul Awal

Department of Computer Science and Operations Research
University of Montreal, QC, Canada
Mila - Quebec AI Institute, QC, Canada

Email: awalrabiul6@gmail.com

Website: rabiul.me

Research Interests: Generative Models, World Models, RL

Current Position

09/2024 – **Mila - Quebec AI Institute, Montreal**
present Graduate Researcher

05/2026 – **Qualcomm Research, Amsterdam**
present Research Intern

Education

09/2024 – **University of Montreal**
present Ph.D. in Computer Science
Department of Computer Science and Operations Research
Advisor: Aishwarya Agrawal
Courses: Reinforcement Learning and Optimal Control, Vision and Language

01/2021 – **University of Saskatchewan**
12/2022 MSc. Computer Science
Advisor: Michael Horsch, Roy Ka-Wei Lee

03/2014 – **Noakhali Science and Technology University, Bangladesh**
12/2018 BSc. Computer Science

Previous Experience

07/2024 – **ServiceNow Research, Montreal**
09/2025 Visiting Researcher in the Multimodal Team
Advisor: Sai Rajeswar

01/2023 – **Mila - Quebec AI Institute, Montreal**
08/2024 Research Intern
Advisor: Aishwarya Agrawal

06/2020 – **Singapore University of Technology and Design**
11/2022 Student Researcher at Social AI Studio
Advisor: Roy Ka-Wei Lee

12/2019 – **Goava Sales Intelligence AB, Dhaka**
12/2020 Software Engineer

In-progress Research

- [4] *Taming Latent Drift in Unified Tokenizer–Denoisers* Mitigates latent-space drift and adversarial instability in joint tokenization–denoising by introducing smoother latent representations, improved training schedules, and a mixture-of-experts design, leading to faster convergence and improved FiD.
- [3] *Dynamics Forcing for Diffusion World Models* Addresses weak temporal dynamics in diffusion video models by borrowing self-supervised learning recipes for predictive latent transitions.
- [2] *Improving Flow-map Training via Feature-space Consistency* Feature space consistency for faster training and improving stability of one-step diffusion models.

Peer-Reviewed Conference Publications

Google Scholar for full list:

<https://scholar.google.com/citations?user=o0cLfUMAAAAAJ&hl=en>

*Denotes equal contribution

- [C8] Aarash Feizi, Shravan Nayak, Xiangru Jian, Kevin Qinghong Lin, Kaixin Li, **Rabiul Awal** + 11 others. Grounding Computer Use Agents on Human Demonstrations. In *International Conference on Learning Representations (ICLR)*. 2026.
- [C7] Juan A. Rodriguez, Haotian Zhang, Abhay Puri, Aarash Feizi, Rishav Pramanik, Pascal Wichmann, Arnab Mondal, Mohammad Reza Samsami, **Rabiul Awal** + 7 others. Rendering-Aware Reinforcement Learning for Vector Graphics Generation. *Conference on Neural Information Processing Systems (NeurIPS)*. 2025.
- [C6] Saba Ahmadi*, **Rabiul Awal***, Ankur Sikarwar*, Amirhossein Kazemnejad*, Ge Ya Luo, Juan A Rodriguez, Sai Rajeswar, Siva Reddy, Christopher Pal, Benno Krojer, Aishwarya Agrawal. The Promise of RL for Autoregressive Image Editing. *Conference on Neural Information Processing Systems (NeurIPS)*. 2025.
Code and models available at <https://github.com/mair-lab/EARL>.
- [C5] Anikter Didolkar*, Andrii Zadaianchuk*, **Rabiul Awal***, Maximilian Seitzer, Efstratios Gavves, Aishwarya Agrawal. CTRL-O: Language-Controllable Object-Centric Visual Representation Learning. *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)*. 2025.
Spotlight presentation at Multimodal Algorithmic Reasoning Workshop, CVPR'25.
- [C4] **Rabiul Awal**, Mahsa Massoud, Aarash Feizi, Zichao Li + 9 others. WebMMU: A Benchmark for Multimodal Multilingual Website Understanding and Code Generation. *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2025.
Spotlight presentation at Multimodal Algorithmic Reasoning Workshop, CVPR.
- [C3] **Rabiul Awal***, Saba Ahmadi*, Le Zhang*, and Aishwarya Agrawal. VisMin: Visual Minimal-Change Understanding. *The Conference on Neural Information Processing Systems (NeurIPS)*. 2024.
Contributed talk at SpLU-RoboNLP Workshop, ACL 2024.
- [C2] Shravan Nayak, Kanishk Jain, **Rabiul Awal**, Siva Reddy, Sjoerd van Steenkiste, Lisa Anne Hendricks, Karolina Stanczak, Aishwarya Agrawal. Benchmarking Vision Language Models for Cultural Understanding. *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2024. **Oral Presentation**.

- [C1] Le Zhang, **Rabiul Awal**, Aishwarya Agrawal. Contrasting Intra-modal and Ranking Cross-modal Hard Negatives to Enhance Visio-linguistic Compositional Understanding. *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)*. 2024.
Spotlight presentation at the Open-Domain Reasoning Multi-Modal Workshop, CVPR'23.

Preprints

- [P1] **Rabiul Awal**, Jinseong Jeong, Ankur Sikarwar, Parisa Kordjamshidi, Andrii Zadaianchuk, Sai Rajeswar, Paul Hongsuck Seo, Aishwarya Agrawal. One Flow-Transformer for Imagination and Control. *preprint*.

Selected Awards, Grant & Fellowship

- 2026 Fonds de recherche du Québec – Nature et technologies (FRQNT) Scholarship (25,000\$).
2026 DIRO Excellence Scholarship for the Winter 2026 session (3900\$)
2025 DIRO Excellence Scholarship for the Fall 2025 session 2025-2026 (3000\$)
2025 Outstanding Reviewer at CVPR
2024 Selected for CIFAR DLRL Summer School at Toronto
2024 Selected for Upper Bound Talent Bursary at Alberta (\$1500)
2022 Selected for Alberta University AI Week Talent Bursary (\$1500)
2022 D&I Awardee at ACL Virtual Conference
2021 Graduate Teaching Fellowship at University of Saskatchewan, Canada
2016 1st place at NSTU Intra University Programming Contest, Dhaka, Bangladesh
2015 3rd place at CoU Regional Programming Contest (CoURPC), Comilla, Bangladesh

Service

Broader Community

- 2024–present **Supervision Request Process Evaluation Committee**
Help professors recruit (pre-evaluation of application) msc and phd students at Mila
- 2025–present **Whiteboard Session Lead Organizer**
Help organize the RL for Foundation Models reading group for internal lab at Mila
- 2021–2022 **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan
- 2023–2024 **Language Grounding Reading Group Organizer**
Help supporting running the reading group on vision language at Mila
- 2021–2022 **Computer Science Graduate Council MSc Representative**
Helped in strengthening graduate research activities, hosting social events for grad students at University of Saskatchewan
- 2021–2023 **NLP Dhaka Reading Group Founder**
A grass-root initiative for fostering ML/NLP research in Bangladesh
Organized ~35 crowd-source reading sessions to spread AI research in local community
- 2017–2018 **NSTU Computing Club President**
Led the computing club growth 10X (from 15 to 150 members), conducted training on competitive programming

Research Community

- 02/2026 **Workshop Organizer at Mila**
World Modeling Workshop
- 05/2025 **Workshop Organizer at CVPR25**
Vision Language Models For All: Building Geo-Diverse and Culturally Aware Vision-Language Models
- 2021–present **Reviewer**
AAAI26, NeurIPS25, CVPR25, ECCV24, CVPR24, ACL23, SIGIR22, ACL22, ICDM21, SIGKDD21
- 2021 **Volunteer**
ICML21, EACL21, ICLR21, NAACL21

Teaching

- 09/2025 – **Mila – Quebec AI Institute, Canada**
12/2025 Teaching Assistant
Computer Science Department
Courses: Representation Learning
- 01/2021 – **University of Saskatchewan, Canada**
12/2022 Graduate Teaching Assistant
Computer Science Department
Courses: Data Structure and Algorithms, Python, Introduction to Artificial Intelligence
- 06/2017 – **Noakhali Science and Technology University, Bangladesh**
06/2018 Teaching Assistant
CSTE 3110 Microprocessor, Micro-controller and Interfacing Lab

Skills

Python (PyTorch, Tensorflow), Java, C/C++, NoSQL (Elasticsearch), AWS (SageMaker, EC2, Lambda)